

AVIRAL NAUTIYAL

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EDUCATION

- M.S. in Computer Science (3.9/4.0)** **Sept. 2024 – May 2026**
Lakehead University *Thunder Bay, ON, Canada*
- **Focus areas:** Machine Learning, Deep Learning, NLP, Data Structures & Algorithms
 - **Master's Research Project:** Brain Tumor Detection from MRI images using Machine Learning
- Master of Computer Application (8.42/10)** **Aug. 2022 – Jul. 2024**
DIT University *Dehradun, Uttarakhand, India*
- **Core Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks
 - **Advanced Coursework:** Java Programming, Software Engineering, Artificial Intelligence, Machine Learning
- Bachelor of Science (Physical Science with Computer Science) (8.9/10)** **Aug. 2019 – Jun. 2022**
University of Delhi *New Delhi, India*
- **Core Coursework:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks
 - **Advanced Coursework:** Java Programming, Software Engineering, Artificial Intelligence, Machine Learning

TECHNICAL SKILLS

Programming Languages & Development: Python (proficient), Java (prior experience), SQL, R(prior experience), Jupyter Notebooks, Object-Oriented Programming (OOP), Data Structures & Algorithms, Latex

Frameworks, Libraries & Project Tools: Scikit-learn, TensorFlow, NumPy, PyTorch, OpenCV, Pandas, Matplotlib, Git, GitHub, GitLab, VS Code, Firebase

Machine Learning & AI: Machine Learning, Deep Learning, Feature Engineering, Transfer Learning, Data Pre-processing, Model Evaluation, NLP, Computer Vision

Platforms: MacOS, Windows, Ubuntu, Linux

PROJECTS

- Brain Tumor Detection from MRI images using Machine Learning** | 🐙 | *Python, TensorFlow, Scikit-learn*
- Developed a brain tumor classification framework using **3 deep feature extractors (ResNet50, VGG16, and EfficientNet-B0)**, generating a **3,840-dimensional fused feature vector** from MRI images.
 - Implemented **Black Hole Optimization (BHO)** feature selection to reduce the feature space from **3,840 features to approximately 2,000 optimal features**, improving computational efficiency while preserving classification performance.
 - Implemented and evaluated an **Optimized Stochastic Gradient Descent based Logistic Regression classifier** across 3 MRI datasets, achieving up to **98.48% classification accuracy** with significantly lower testing time compared to traditional machine learning models.
- Traffic Sign Recognition System** | 🐙 | *Python, TensorFlow, OpenCV*
- Developed a **CNN-based multi-class image classification system** to recognize 43 traffic sign classes using the GTSRB dataset.
 - Implemented image preprocessing pipelines on **50,000+ labelled traffic sign images** including normalization, resizing, and data augmentation techniques.
 - Designed and trained deep CNN architectures with regularization techniques, achieving **approximately 97% classification accuracy** on unseen test data.
 - Implemented structured training and validation workflows with early stopping, **converging within 20–30 epochs**.
- AI-Based Facial Recognition Attendance System** | 🐙 | *Python, OpenCV, Firebase*
- Designed and implemented a **real-time computer vision system** for facial recognition-based attendance tracking.
 - Built pipelines for face detection, feature encoding, and similarity matching using a dataset of 100 labelled face images.
 - Integrated the application with **Firebase cloud-based storage and databases** for persistent attendance records and student metadata.
 - Achieved **approximately 80–85% recognition accuracy** under controlled lighting conditions during system evaluation.